EXPT-2: MEASUREMENT OF MAGNETIC FLUX (Deflection and Vibration Magnetometer)

AIM:

To determine the horizontal component of earths magnetice flux density at the place using deflection and vibration magnetometers

APPARATUS:

Deflection magnetometer, box type vibration magnetometer, stop watch, magnet etc

The box type vibration magnetometer consists of a rectangular box provided with glass sides. A stirrup is suspended by a torsionless fibre, from the top of a vertical tube fixed centrally on the top side of the box, through which oscillations of the magnet can be observed.

PRINCIPLE:

The period of oscillation of the bar magnet

$$T = 2\pi\sqrt{\frac{I}{mB_h}}$$

Where I is the moment of inertia of the magnet about the axis of oscillation, m= dipole moment of the magnet and B_h = Earth's horizontal flux density at the place

Therefore

$$T^2 = 4\pi^2 \frac{I}{mB_h} \quad \text{or} \quad mB_h = \frac{4\pi^2 I}{T^2}$$

$$I = Mass\,of\,the\,magnet \times \frac{Length^2 + Breadth^2}{12}$$

Length of the magnet = $2l$

Using a deflection magnetometer adjusted for tan A position,

$$\frac{\mu_0}{4\pi}\frac{2md}{\left(d^2 - l^2\right)^2} = B_h \tan\theta$$

$$\frac{m}{B_h} = \frac{2\pi\left(d^2 - l^2\right)^2 \tan\theta}{\mu_0 d}$$

$\mu_0 = 4\pi X10^{-7}\ Henry/m$

In tan B position,

$$\frac{m}{B_h} = \frac{4\pi}{\mu_0}\left(d^2 + l^2\right)^{3/2} \tan\theta$$

Let $mB_h = X$ and

$$\frac{m}{B_h} = Y$$

$$\frac{mB_h}{\frac{m}{B_h}} = \frac{X}{Y}$$

$$B_h^2 = \frac{X}{Y} \quad \text{or}$$

$$B_h = \sqrt{\frac{X}{Y}}$$

PROCEDURE:

i. Determination of X(= mB_h)

The bar magnet is placed horizontally in the stirrup of the vibration magnetometer, with its north pole pointing north. The box is rotated so that its length is parallel to the magnet(ie along the magnetic meridian). Another magnet is bought near the vibration magnetometer and suddenly withdrawn so that the magnet inside the box begins to oscillate. The time for 10 oscillations is determined twice or thrice. Hence the period of oscillation (T) is calculated

The mass of magnet is determined. Its length and breath are measured using vernier calipers and its moments of inertia about the axis of oscillation are calculated. Hence the value of mB_h is calculated as $mB_h = \frac{4\pi^2 I}{T^2}$. Let it be X

ii Determination of Y(= $\frac{m}{B_h}$)

The deflection magnetometer is arranged in the **tan A position**.(For this thecompass box alone is rotated so that the (0-0) line is parallel to the arm of the magnetometer. Then the deflection magnetometer apparatus as a whole is rotated till the aluminium pointer reads (0-0). **Now the magnet in the compass box is perpendicular to the arm.)** The bar magnet (length parallel to the arm) is placed with its center at a distance (d) from the center of compass box. The mean of eight deflections is determined as θ.(the eight defelctions are obtained in the compass box by keeping magnet on one arm of the deflection magnetometer, then by reversing the magnet poles at the same position, and then by repeating the same thing at the same distance as on the other arm) the length $2l$ of the magnet is measured. Hence $\frac{m}{B_h}$ is calculated. The experiment is repeated for different values of d and mean value of $\frac{m}{B_h}$ is determined.

$\frac{m}{B_h}$ may also be determined by arranging the deflection magnetometer in tan B position[For adjustment in the tan B position, the compass box alone is rotated until the line joining(90-90) readings is parallel to the scale. Then the deflection magnetometer as a whole is rotated until the pointer reads (0-0). **Now the magnet in the compass box is parallel to the arm.** The bar magnet (length perpendicular to the arm) is placed with its center at a distance (d) from the center of compass box. The mean of eight deflections is determined as θ. The mean value of $\frac{m}{B_h}$ obtained by tan A and tan B methods is determined. Let this mean value be Y.

OBSERVATIONS:

i To find mB_h

Mass of the magnet = gm = Kg

Length & Breadth of the magnet

Value of one main scale division =
Number of divisions on the vernier scale n =
Least count = value of one main scale division ÷ n =

Trial No	M.S.R cm	V.S.R	LENGTH=M.S.R +(V.S.R x LC.) cm

Mean Length =.....m

Trial No	M.S.R	V.S.R	BREADTH=M.S.R +(V.S.R x LC.)

Mean Breadth =.....m

Time period of oscillations

Time for 10 oscillations (sec)				Period T sec
1	2	3	Mean	

$$I = Mass\ of\ the\ magnet \times \frac{Length^2 + Breadth^2}{12} =$$

$= \ldots\ldots\ldots Kgm^2$

mB_h = X $= \frac{4\pi^2 I}{T^2}$ =

ii To find ($\frac{m}{B_H}$)

Length of the magnet 2*l* =.....m

Trial No	Distance d (metre)	Deflections								Mean θ	$\frac{m}{B_h}$
		1	2	3	4	5	6	7	8		
Tan A											
Tan B											

Mean value of $\frac{M}{B_h}$ = Y =

Horizontal intensity B_h $= \sqrt{\frac{x}{y}} =$

Moment of the magnet $= \sqrt{xy} =$.

RESULT:

The horizontal component of earth's magnetic flux density = Tesla
Moment of the given magnet = $A\text{-}m^2$